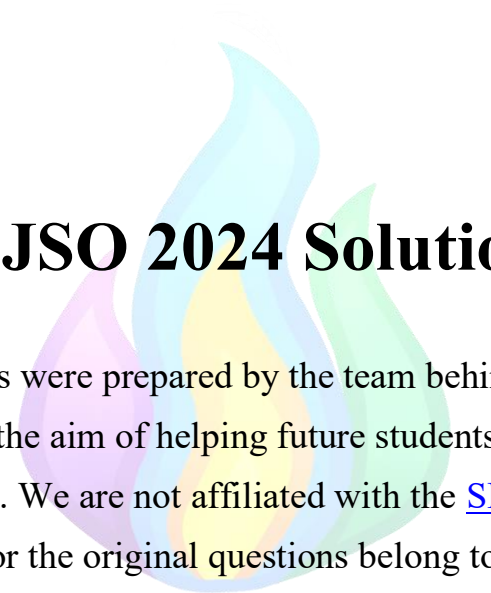




SLJSO 2024 Solutions

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Question No.	Field	References
<u>1</u>	Human Biology	<u>Animal Tissues</u>
<u>2</u>	Cell Biology	<u>Biomolecules</u>
<u>3</u>	Cell Biology	<u>Biomolecules</u>
<u>4</u>	Molecular Biology	<u>Structure of DNA</u>
<u>5</u>	Cell Biology	<u>Prokaryotic Cells</u>
<u>6</u>	Cell Biology	<u>Cell Organelles</u>
<u>7</u>	Cell Biology	<u>Meiosis</u>
<u>8</u>	Cell Biology	<u>Cellular Respiration</u>
<u>9</u>	Botany	<u>Photosynthesis</u>
<u>10</u>	Human Biology	<u>Circulatory System</u>
<u>11</u>	Zoology	<u>Osmoregulation</u>
<u>12</u>	Botany	<u>Transport in Plants</u>
<u>13</u>	Zoology	<u>Osmoregulation</u>
<u>14</u>	Ecology	<u>Food Chains and Food Webs</u>
<u>15</u>	Ecology	<u>Ecological Pyramids</u>
<u>16</u>	Cell Biology	<u>Biomolecules</u>
<u>17</u>	Chemical Kinetics	<u>Rate Law</u>
<u>18</u>	Chemical Structures	<u>Trends in the Periodic Table</u>
<u>19</u>	Electrochemistry	<u>Electrochemical Potential</u>
<u>20</u>	Chemical Kinetics	<u>Rate Law</u>
<u>21</u>		
<u>22</u>		
<u>23</u>	Chemical Equilibrium	<u>Reaction Stoichiometry</u> <u>pH Scale</u>
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<u>25</u>	Chemical Equilibrium	<u>Conjugate Bases</u>
<u>26</u>	Electrochemistry	<u>Electrochemical Cells</u>

<u>27</u>	Stoichiometry	<u>Concentration</u> <u>Reaction Stoichiometry</u>
<u>28</u>	Electrochemistry	<u>Electrochemical Cells</u>
<u>29</u>	Electrochemistry	<u>Redox Reactions</u>
<u>30</u>	Thermochemistry	<u>Gibbs Free Energy</u>
<u>31</u>	Thermodynamics	<u>Specific Heat Capacity</u>
<u>32</u>	Electromagnetism	<u>Kirchoff's Laws</u> <u>Electric Power</u>
<u>33</u>	Mechanics	<u>Kinematics Formulas</u>
<u>34</u>	Mechanics	<u>Work-Energy Relationship</u>
<u>35</u>	Mechanics	<u>Conservation of Momentum</u>
<u>36</u>	Thermodynamics	<u>Ideal Gas Law</u>
<u>37</u>	Light	<u>Refraction</u>
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<u>39</u>	Electromagnetism	<u>Kirchhoff's Laws</u>
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<u>43</u>	Mechanics	<u>Torque</u> <u>Newton's Laws</u>
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<u>45</u>	Electromagnetism	<u>Transformers</u>
<u>46</u>	Thermodynamics	<u>Specific Heat Capacity</u>
<u>47</u>	Algebra	
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<u>50</u>	Algebra	

Question 1

Molecules, Organelles, Cells, Tissues, Organs, Organ systems, Organism. This sequence shows the correct hierarchy of complexity in living organisms, starting from basic molecules to a mature organism.

Correct Answer: C

Question 2

Water has a high specific heat capacity. The large specific heat of water enables it to absorb or release great quantities of heat without a large temperature change, helping organisms like us maintain stable internal conditions.

Correct Answer: C

Question 3

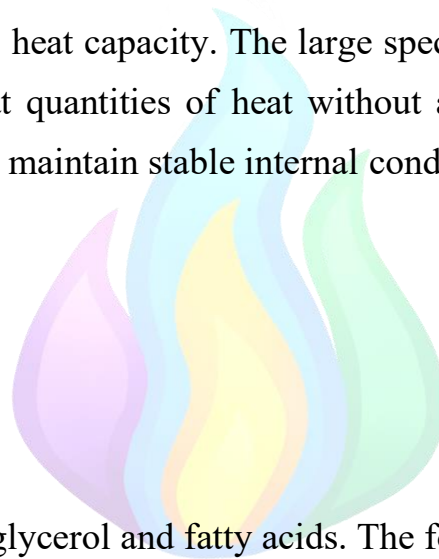
Fats are hydrolyzed into glycerol and fatty acids. The formula given corresponds to glycerol, confirming that fat is the correct parent compound.

Correct Answer: B

Question 4

If 20% are adenine, 20% are thymine. The remaining 60% are split equally between guanine and cytosine, resulting in 3000 cytosine bases.

Correct Answer: B



Question 5

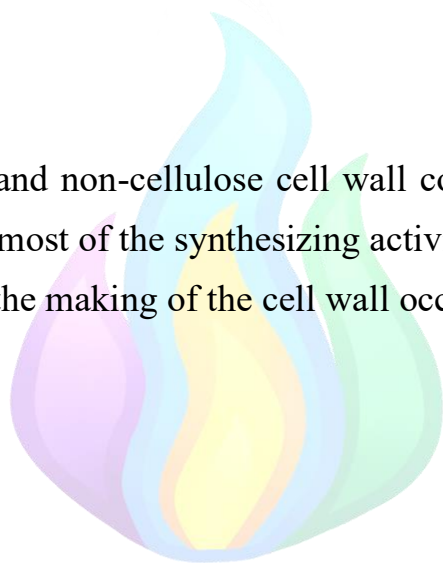
All prokaryotic organisms do not have peptidoglycan in their cell walls. Some prokaryotes like archaea lack peptidoglycan in the cell wall, while others, bacteria have it. Thus, not all prokaryotes are made up of peptidoglycan.

Correct Answer: B

Question 6

Manufacturing cellulose and non-cellulose cell wall components. The plant cell's Golgi apparatus is where most of the synthesizing activities like cellulose and other components required for the making of the cell wall occur.

Correct Answer: B



Question 7

Four daughter cells with the same number of chromosomes are produced. Meiosis creates four genetically unique cells, with half the original cell quantity of chromosomes, ensuring variation in sexual reproduction.

Correct Answer: B

Question 8

Conversion of starch to sugar during seed germination. This process is driven by enzymes and does not directly use ATP, unlike biosynthetic pathways that depend drastically on energy input.

Correct Answer: B

Question 9

Light energy is converted to chemical energy. Photosynthesis captures sunlight to produce glucose and other carbohydrates, storing the energy in chemical bonds.

Correct Answer: D

Question 10

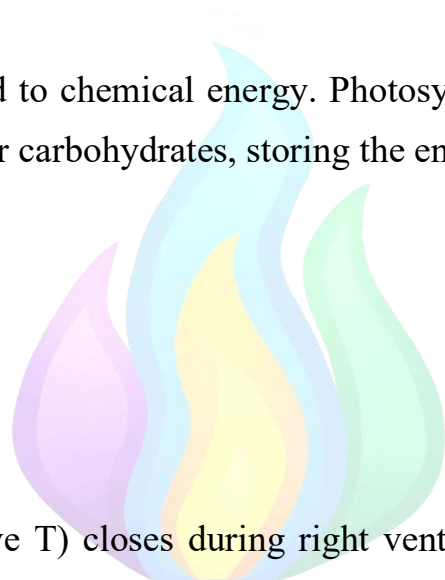
The tricuspid valve (valve T) closes during right ventricle contraction to prevent backflow into the atrium, thus blood moves to the lungs.

Correct Answer: C

Question 11

Solute potential and pressure potential of cell sap have equal and opposite values. On turgidity, pressure potential equilibrates the solute potential exactly, resulting in no net water movement.

Correct Answer: C



Question 12

Transport of food material can occur in both directions within the phloem. (III) and Transport of food material is stopped when phloem tissue is treated with respiratory inhibitors. (IV).

Correct Answer: A

Question 13

Statements III and IV are incorrect because the calculated pressure potentials of the cells do not match the given values.

The pressure potential of cell A is 3.6 MPa, not 1.0 MPa as stated in III. The pressure Potential in cell B is 0.4 MPa, not 0.6 MPa as stated in IV.

Correct Answer: C

Question 14

B may be a lizard, and C may be a snail. This statement accurately describes possible roles in a food web, snails often act as herbivores, and lizards as carnivorous secondary consumers.

Correct Answer: D

Question 15

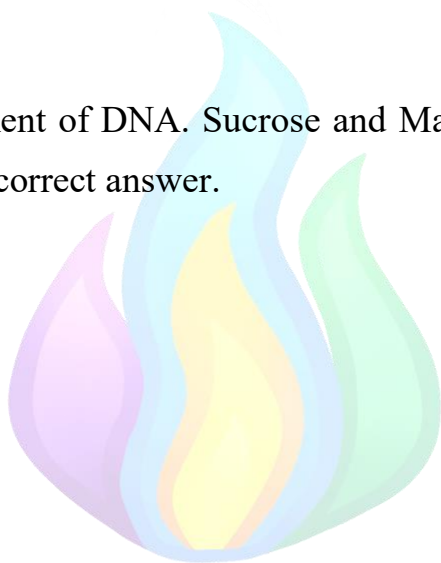
From gross primary productivity (2000 kJ), we can estimate net productivity at the first level using trophic efficiency. Subtracting net from gross gives the energy lost to respiration, 1100 kJ.

Correct Answer: C

Question 16

Deoxyribose is a component of DNA. Sucrose and Maltose are disaccharides. So, option D (Lactose) is the correct answer.

Correct Answer: D



Question 17

In a question like this, it is obvious that the rate of the overall reaction is equal to the rate of the slow step.

So, we just must apply the rate law to the slow step.

$lA + mB \Rightarrow nC + oD$ where l , m , n , and o are the coefficients of the balanced reaction, then

The rate of the reaction = $k[A]^l[B]^m$ where k represents the rate constant

Applying this to the above reaction, we get the rate of the reaction = $k_1[NO_2]^2$

Correct Answer: D

Question 18

When we move from left to right in the p-block, the number of protons and the number of electrons increase. This means the Coulomb attraction between the protons and neutrons increases. Due to this, the atomic radius decreases gradually.

Correct Answer: D

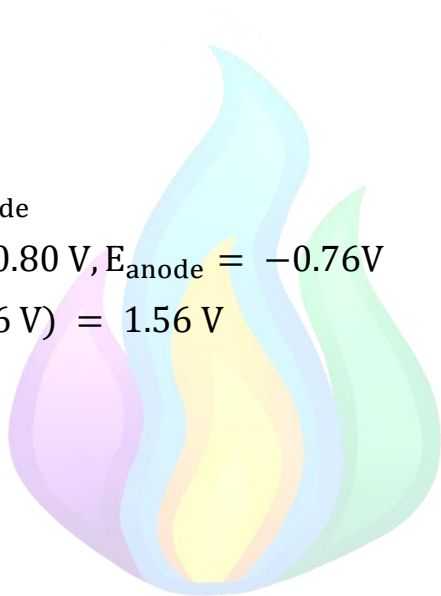
Question 19

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$\text{Substituting } E_{\text{cathode}} = 0.80 \text{ V}, E_{\text{anode}} = -0.76 \text{ V}$$

$$E_{\text{cell}} = 0.80 \text{ V} - (-0.76 \text{ V}) = 1.56 \text{ V}$$

Correct Answer: A



Question 20

A first-order reaction means the rate of the reaction = $k [\text{Reactant}]^1$

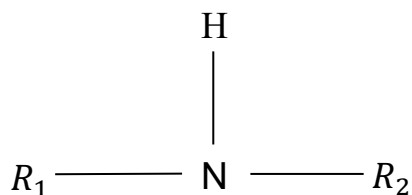
Substituting values, we get $0.6 \text{ Ms}^{-1} = 0.035 \text{ s}^{-1} [\text{Reactant}]$

Hence, $[\text{Reactant}] = 17.143 \text{ M}$

Correct Answer: B

Question 21

A secondary amine is an amine with two alkyl groups attached to it.



- Option A includes two R groups (CH_3CH_2 and CH_3) bonded to the N atom. So, it is a secondary amine.
- Option B includes three R groups (CH_3 , CH_3 , CH_3) bonded to the N atom. So, it is a tertiary amine.
- Option C includes three R groups ($\text{CH}_3\text{CH}_2\text{CH}_2$, $\text{CH}_3\text{CH}_2\text{CH}_2$, CH_3) bonded to the N atom. So, it is a tertiary amine.
- Option D includes a C_6H_5 group bonded to the N atom. This compound is called aniline and is a primary amine.

Correct Answer: A

Question 22

- (A) It gives blue color with iodine—true, starch is used as an indicator in iodometric titrations, turning blue in the presence of iodine.
- (B) It is a polymer of α -D-glucose—true, starch consists of two polysaccharides, amylose and amylopectin, both of which are α -D-glucose polymers.
- (C) It is a reducing carbohydrate—false, starch is not a reducing carbohydrate. Although one might think it reduces molecular iodine to its -1-oxidation state, it just forms a complex species in which iodine is physically adsorbed in the structure of the polymer (specifically amylose).
- (D) It consists of branched chains—true, one of the polysaccharides in the structure of starch, specifically amylopectin, has a branched structure.

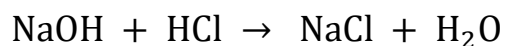
Correct Answer: C

Question 23

Assume there is a V volume of NaOH and a V volume of HCl.

$$\text{No. of NaOH moles} = C \times V = 0.1 \times V = 0.1V$$

$$\text{No. of HCl moles} = C \times V = 0.01 \times V = 0.01V$$



The 0.01V number of HCl moles reacts with the 0.01V number of NaOH moles.

This leaves a solution that consists of 0.09 moles of NaOH and a volume of 2V.

$$\text{Concentration} = \frac{\text{No. of Moles}}{\text{Volume}} = \frac{0.09V}{2V} = 0.045$$

$$\text{pOH} = -\log [\text{OH}^-] = -\log [\text{NaOH}] = -\log [0.045] = 1.35$$

$$\text{pH} = 14 - \text{pOH} = 14 - 1.35 = 12.65$$

Correct Answer: C

Question 24

There are 3 stages of addition polymerization.

1. Initiation—A free radical or a catalyst starts the reaction. The initiator breaks a double bond in the monomer, creating an active site.
2. Propagation—The active monomer radical reacts with another monomer, extending the polymer chain. The process continues as new monomers keep adding to the chain.
3. Termination—The reaction stops when two growing chains combine or when a radical reacts with an inhibitor. This results in a stable polymer with no further chain growth.

Correct Answer: B

Question 25

The strongest conjugate base is the weakest acid. Out of the 4 options, it is obvious that the weakest acid is CH_3COO^- .

Correct Answer: C

Question 26

- (A) - False, in the cathode, reactants gain electrons during the reaction. Loss of electrons means reduction.
- (B) - True, in the anode, electrons are lost during the reaction. These electrons move towards the cathode.
- (C) - False, the cathode is denoted by a positive sign.
- (D) - False, it is usually made up of conducting material'

Correct Answer: B

Question 27

$$\text{H}^+ \text{ moles} = C \times V = 0.05\text{M} \times \frac{20}{1000}\text{L} = 0.001 \text{ moles}$$

$$\text{OH}^- \text{ moles} = C \times V = 2 \times C \times V = 2 \times 0.10\text{M} \times \frac{30}{1000}\text{L} = 0.006 \text{ moles}$$

$$\text{OH}^- \text{ moles remaining} = 0.006 \text{ moles} - 0.001 \text{ moles} = 0.005 \text{ moles}$$

$$\text{Total Volume} = 20\text{mL} + 30\text{mL} = 50\text{mL}$$

$$\text{Concentration of } [\text{OH}^-] = \frac{\text{No. of Moles}}{\text{Volume}} = \frac{0.005 \text{ moles}}{50/1000 \text{ L}} = 0.10\text{M}$$

Correct Answer: C

Question 28

Copper has a higher (more positive) reduction potential than iron. The electrode with a higher reduction potential will be the cathode. Hence, copper will be the cathode, and iron will be the anode.

Correct Answer: B

Question 29

N_2O_3 has a net oxidation state of zero. The oxidation state of O is -2. So, the three oxygen atoms will have an oxidation number of -6. So, the two nitrogen atoms should have an oxidation state of +6. This means each nitrogen atom should have an oxidation state of +3.

Correct Answer: C

Question 30

$$\Delta G = \Delta H - T\Delta S$$

For a reaction to be feasible, the Gibbs free energy should be less than zero (negative).

Correct Answer: B

Question 31

We all know the specific heat capacity formula, $Q = mc \Delta T$

For material A, substituting $Q = 1000 \text{ J}$, $m = 0.75 \text{ kg}$, and $c = 4200 \text{ J/kg.K}$

$$\Delta T = \frac{Q}{mc} = \frac{1000\text{J}}{0.75\text{kg} * 4200\text{J/kg.K}} = 0.317 \text{ K}$$

For material B, substituting $Q = 1000 \text{ J}$, $m = 1.00 \text{ kg}$, and $c = 2400 \text{ J/kg.K}$

$$\Delta T = \frac{Q}{mc} = \frac{1000\text{J}}{1\text{kg} * 2400\text{J/kg.K}} = 0.417 \text{ K}$$

For material C, substituting $Q = 1000 \text{ J}$, $m = 0.50 \text{ kg}$, and $c = 2500 \text{ J/kg.K}$

$$\Delta T = \frac{Q}{mc} = \frac{1000\text{J}}{0.5\text{kg} * 2500\text{J/kg.K}} = 0.800 \text{ K}$$

For material D, substituting $Q = 1000 \text{ J}$, $m = 1.25 \text{ kg}$, and $c = 2100 \text{ J/kg.K}$

$$\Delta T = \frac{Q}{mc} = \frac{1000\text{J}}{1.25\text{kg} * 2100\text{J/kg.K}} = 0.381 \text{ K}$$

In conclusion, material C will have the highest temperature change.

Correct Answer: C

Question 32

To solve this question, we would have to use $P = I^2 R$ and substitute $P = 25 \text{ W}$ and $R = 100 \Omega$.

$$I = \sqrt{\frac{P}{R}} = \sqrt{\frac{25 \text{ W}}{100 \Omega}} = 0.5 \text{ A}$$

The maximum possible current through any resistor is 0.5A. This means the current through the leftmost resistor must be $\leq 0.5 \text{ A}$.

If 0.5A flows through the leftmost resistor, the current splits equally, with 0.25A flowing through each of the resistors on the right. As a result, the power dissipated in those resistors will be less than 25 W.

Thus, the correct answer is 0.5A.

Hint: Note that some students take the current flowing through each of the resistors on the right to be 0.5 A and write the total resistance to be 1.0 A. However, if a current of 1.0 A flows through the resistor on the left, the power dissipated will be 100 W, which is higher than 25 W.

Correct Answer: B

Question 33

To find the time elapsed, we need to find the deceleration of the bicycle. We are given the initial velocity, the final velocity, and the distance traveled.

To solve for acceleration, we will use the fundamental equation $v^2 = u^2 + 2as$

Rearranging, we get $a = \frac{v^2 - u^2}{2s}$

Substituting $v = 0\text{ms}^{-1}$, $u = 10\text{ms}^{-1}$, $s = 20\text{m}$ into the above equation, we get

$$a = \frac{0^2 - 10^2}{2 * 20} = -2.5\text{m/s}^2$$

Then, we will use the equation $v = u + at$ to solve for the time elapsed.

Rearranging, we get $t = \frac{v - u}{a}$

Substituting $v = 0\text{ms}^{-1}$, $u = 10\text{ms}^{-1}$, $a = -2.5\text{ms}^{-2}$ into the above equation, we get

$$t = \frac{0 - 10}{-2.5} = 4\text{s}$$

Correct Answer: B

Question 34

Work done is defined as the **energy transferred** to or from an object when a force is applied over a displacement. The only answer that fits the definition is option B.

Correct Answer: B

Question 35

To solve problems like this one to find the total final energy, we need to find the final velocities. So, we must apply the conservation of momentum.

Initial Momentum = Final Momentum

$$\text{Initial Momentum} = m_s v_s + m_w v_w = 1\text{kg} \times 0 + 0.5\text{kg} \times 6\text{ms}^{-1} = 3\text{kgms}^{-1}$$

$$\text{Hence, the final momentum} = 3\text{kgms}^{-1}$$

$$\text{Final momentum} = (m_s + m_w) v = 3\text{kgms}^{-1} = (1\text{kg} + 0.5\text{kg}) v = 1.5v$$

Hence, the blocks stick together and travel with a velocity of 2 m/s.

Now, we shall find the final total energy of the system.

$$\text{Final Total Energy} = \frac{1}{2}(m_s + m_w)v^2 = \frac{1}{2} (1\text{kg} + 0.5\text{kg}) * (2\text{ms}^{-1})^2 = 3\text{J}$$

Correct Answer: A

Question 36

To find the number of atoms, we must use the ideal gas law, $Pv = nRT$

Rearranging, we get Moles = $\frac{Pv}{RT}$

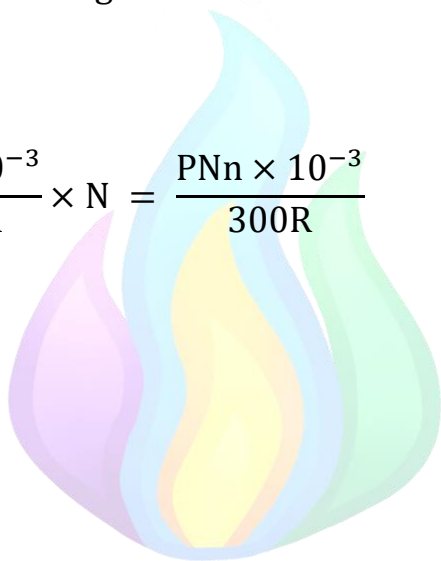
Substituting $n \times 10^{-3} = v$, $T = (27 + 273)K = 300K$, we get Moles = $\frac{Pn \times 10^{-3}}{300R}$

No. of Atoms = Moles $\times$ Avogadro Constant

Substituting, we get

$$\text{No. of Atoms} = \frac{Pn \times 10^{-3}}{300R} \times N = \frac{PNn \times 10^{-3}}{300R}$$

Correct Answer: C



Question 37

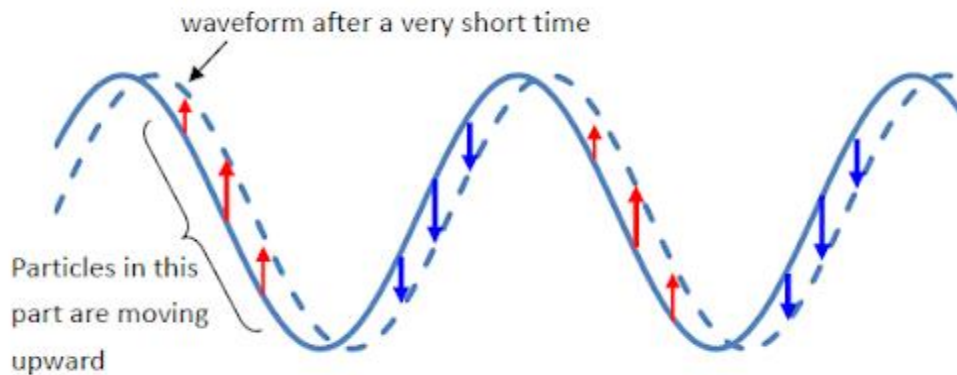
Several fundamental concepts of light are tested in this problem. You need to have a clear understanding of refraction and how light travels.

- Light always travels in a straight line when the refractive index is the same. So, option C can be eliminated.
- According to Snell's law, light bends away from the normal when travelling from a material with a higher refractive index (in this case, water) to a material with a lower refractive index (in this case, air). So, options A and D can be eliminated.

Correct Answer: B

Question 38

To solve this question, you must be familiar with how transverse waves travel through a surface and how they affect a particle on the surface.



As you can see, the particles in the surface vibrate up and down the surface. So, choice C can be eliminated. You can also observe that the particle's position relative to the surface does not change. So, options A and B can be eliminated. Choice D states that the particles vibrate up and down while staying in the same position. Hence, choice D is the correct answer.

Correct Answer: D

Question 39

Assume the voltage of the battery to be V and the resistor of each bulb to be R . The voltage through each line of resistors is also V .

In each line of resistors, the voltage is equally distributed among each resistor. So, the voltage through resistors $L1$ and $L2$ is $V/2$ each, the voltage through resistor $L3$ is V , and the voltage through the resistors $L4$, $L5$ and $L6$ is $V/3$ each.

The power through a resistor is given by the equation $E = I^2R = VI = \frac{V^2}{R}$.

So, the power through $L1$, $L2$, $L3$, $L4$, $L5$ and $L6$ is $\frac{(V/2)^2}{R}, \frac{(V/2)^2}{R}, \frac{V^2}{R}, \frac{(V/3)^2}{R}, \frac{(V/3)^2}{R}, \frac{(V/3)^2}{R}$ respectively

So, $L3$ will be the brightest while $L4$, $L5$ and $L6$ will have the minimum brightness.

Hence, only statement II is correct.

Correct Answer: B

Question 40

Torque = Force x perpendicular distance

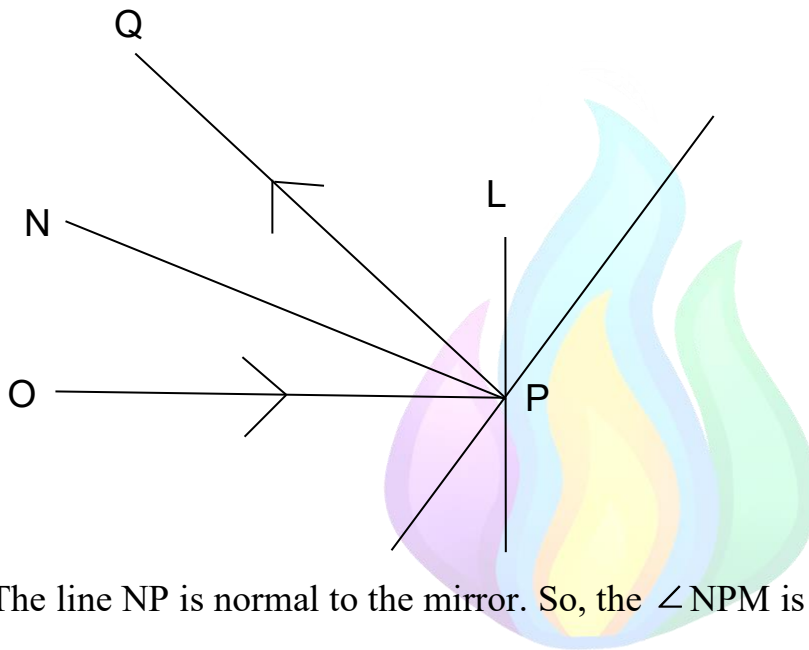
For the plank to stay in equilibrium, the net moment acting around the pivot should be 0. When the force F increases, the anticlockwise moment increases. So, the clockwise moment must be increased.

- In option A, when we decrease the force Q, the clockwise moment decreases. So, the plank will have a net anticlockwise moment. Option A is wrong.
- In option B, when we increase the force P, the anticlockwise moment increases further. So, the plank will have a net anticlockwise moment. Option B is wrong.
- In option C, when we increase the perpendicular distance between force P and the pivot, the anticlockwise moment increases. So, the plank will have a net anticlockwise moment. Option C is wrong.
- In option D, when we increase the perpendicular distance between force Q and the pivot, the clockwise moment increases. This could result in a net zero moment. So, option D is the only possible answer.

Correct Answer: D

Question 41

This question can be solved very easily using a little knowledge of geometry and the law of reflection. The law of reflection states that the incident angle is equal to the reflected angle.



The line NP is normal to the mirror. So, the $\angle NPM$ is 90 degrees. Since $\angle LPM$ is 15 degrees and $\angle LPO$ is 90 degrees. Hence, $\angle OPN$ is 15 degrees.

According to the law of reflection, $\angle OPN$ is equal to $\angle NPQ = 15$ degrees.

Hence, $\angle OPQ = 15 \text{ degrees} + 15 \text{ degrees} = 30 \text{ degrees}$.

Correct Answer: D

Question 42

Assume the original distance between the student and the wall is x .

Time taken to hear the echo = 2 x time taken for the sound to reach the wall =

$$2 \times \frac{\text{distance}}{\text{speed}}$$

$$\text{Time taken to hear the echo} = t_1 = \frac{2x}{v_s} \text{ - equation 1}$$

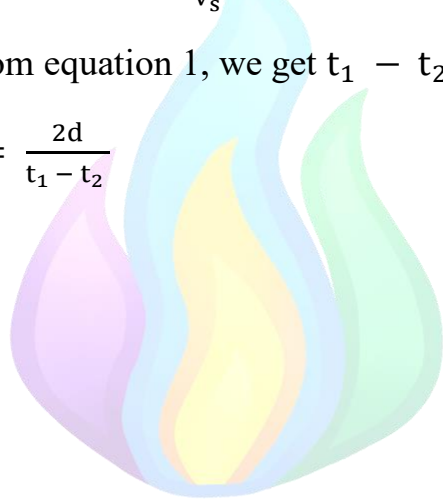
After moving a distance d towards the wall, the new distance is $x-d$.

$$\text{Time taken to hear the echo} = t_2 = \frac{2(x-d)}{v_s} \text{ - equation 2}$$

$$\text{Subtracting equation 2 from equation 1, we get } t_1 - t_2 = \frac{2x - 2(x-d)}{v_s} = \frac{2d}{v_s}$$

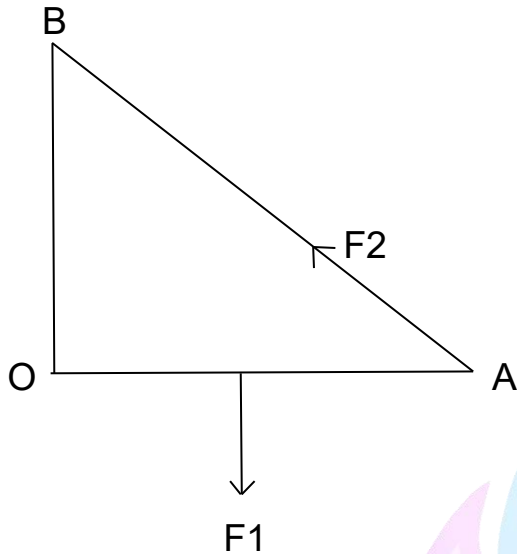
$$\text{Rearranging, we get } v_s = \frac{2d}{t_1 - t_2}$$

Correct Answer: A



Question 43

First, we will analyze what is happening here in the problem.



Force F_1 generates a clockwise moment around point O. For the system to be in equilibrium, force F_2 should generate a similar but counterclockwise moment around point O.

$F_1 = \text{weight of the rod} + \text{weight of the flowerpot}$

$F_2 = \text{Tension}$

So, the correct choice is B.

Correct Answer: B

Question 44

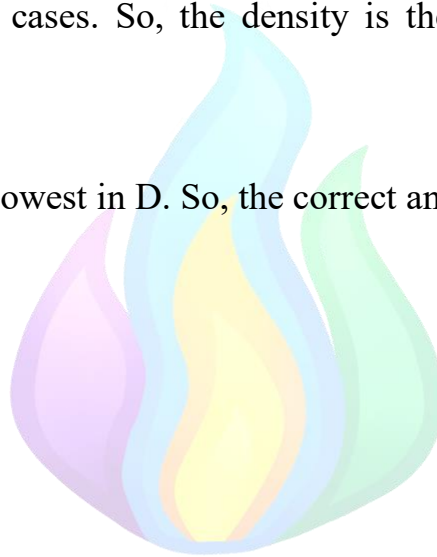
Upthrust (also known as buoyant force), $U = Vdg$ where V represents the volume immersed, d represents the density, and g represents gravity.

According to the equilibrium of forces, $\text{Weight} = mg = \text{Upthrust}$

Weight is a constant force and is always equal to the upthrust. This implies that the upthrust is similar in all cases. So, the density is the highest when the volume immersed is the lowest.

Volume immersed is the lowest in D. So, the correct answer is D.

Correct Answer: D



Question 45

The voltage in the secondary coil is less than that of the primary coil. Such transformers are called step-down transformers.

In a transformer, $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ where V_s is the voltage on the secondary coil, V_p is the voltage on the primary coil, N_s is the number of turns on the secondary coil, and N_p is the number of turns on the primary coil.

Substituting $V_s = 12V$, $V_p = 240V$, and $N_p = 600$ into the above equation, we obtain N_s to be 30.

Correct Answer: C



Question 46

Assume Q heat is supplied in one minute. $Q = mc\Delta T \Rightarrow c = \frac{Q}{m\Delta T}$

Substituting $Q = x$, $\Delta T = 2K$, we get

$$c = \frac{x}{2m} - \text{Equation 1}$$

$$Q = mL \Rightarrow L = \frac{Q}{m}$$

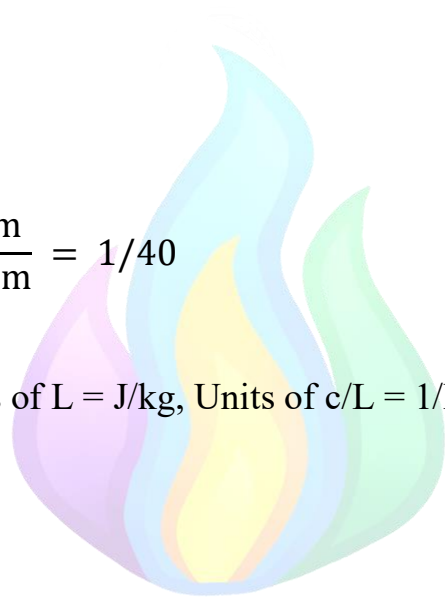
The heater was running for 20 min. So, Q equals 20x. Substituting 20x into Q, we get

$$L = \frac{20x}{m} - \text{Equation 2}$$

$$\frac{\text{Equation 1}}{\text{Equation 2}} = \frac{c}{L} = \frac{x/2m}{20x/m} = 1/40$$

Units of $c = J/kg.K$, Units of $L = J/kg$, Units of $c/L = 1/K$

Correct Answer: A



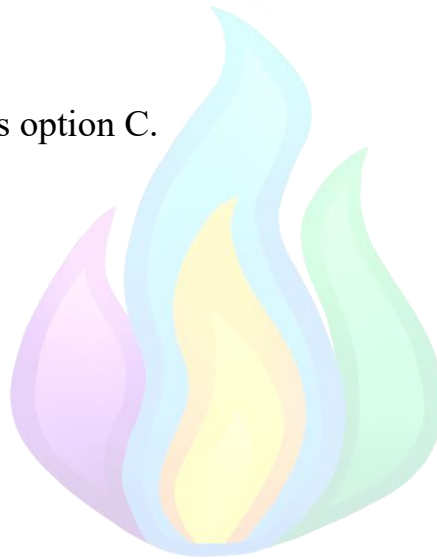
Question 47

If two lines are parallel, then their gradient must be the same. In this case, the gradient must be equal to $10/2 = 5$. So, we can eliminate option B.

- In Option A, when $x=0$, $2y=6$, implying $y=3$. The y-intercept is 3. So, option A is wrong.
- In Option C, when $x=0$, $y=6$. The y-intercept is 6. So, option C is correct.
- In Option D, when $x=0$, $2y=-6$, implying $y=-3$. The y-intercept is -3. So, option D is wrong.

The only correct answer is option C.

Correct Answer: C



Question 48

$$\text{Left hand side} = \frac{4^{2x+2}}{8} = \frac{2^{4x+4}}{2^3} = 2^{4x+1}$$

$$2^{4x+1} = 2^{3x-1}$$

$$4x + 1 = 3x - 1$$

$$x = -2$$

Correct Answer: B

Question 49

Assume Janasha's age is x . Then, Sohani's age is $x+16$.

In 5 years, Janasha will be $x+5$ and Sohani will be $x+21$.

$$\text{If } x + x+5 = \frac{1}{2} (x + 21)$$

Solving, we get $X = 11$.

Janasha is 11 years old now.

Correct Answer: B

Question 50

$$\frac{Q}{t} = \frac{P\pi a^4}{8\ln} + V \Rightarrow \frac{Q}{t} - V = \frac{Q - Vt}{t} = \frac{P\pi a^4}{8\ln}$$

$$\frac{Q - Vt}{t} \times \frac{8\ln}{P\pi} = a^4$$

$$\left(\frac{Q - Vt}{t} \times \frac{8\ln}{P\pi} \right)^{1/4} = a$$

Correct Answer: B